

Slope to Derivative Practice

GRADE 11 · 30 QUESTIONS

Ten questions at each level; work up through the levels. Page 2 holds the solutions, so page 1 can be printed alone for students. Mashqlar uch darajada · Задания трёх уровней.

SLOPE OF A LINE	SECANT SLOPE	DERIVATIVE
$\frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$	$\frac{f(x+h) - f(x)}{h}$	$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

●○○ Section A · Easy

Oson · Лёгкие

One step each — read the numbers straight off the formula.

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| <p>A1 Find the slope of the line through A(1, 2) and B(5, 10).</p> <p>A2 A line has slope 3 and passes through (0, -1). What is y when x = 2?</p> <p>A3 Find the gradient of the line through (-2, 5) and (4, 5).</p> <p>A4 For $f(x) = x^2$, find the slope of the secant joining $x = 1$ and $x = 4$.</p> <p>A5 For $f(x) = x^2$, find the slope of the secant joining $x = 2$ and $x = 5$.</p> | <p>A6 For $f(x) = 3x + 1$, find the secant slope between $x = 1$ and $x = 6$. What do you notice?</p> <p>A7 Given $f(x) = 2x$, find the gradient of the tangent to $y = x^2$ at $x = 3$.</p> <p>A8 Given $f(x) = 2x$, where is the tangent to $y = x^2$ horizontal?</p> <p>A9 For $f(x) = x^2$, simplify $\frac{f(2+h) - f(2)}{h}$.</p> <p>A10 A curve has $f'(4) = 7$. Say in words what this means.</p> |
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●●○ Section B · Medium

O'rta · Средние

Two or three steps: set it up, simplify, then let $h \rightarrow 0$.

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| <p>B1 Find the equation of the line through (2, 3) and (6, 11).</p> <p>B2 For $f(x) = x^2$, find the secant slope between $x = 3$ and $x = 3 + h$.</p> <p>B3 Use the definition of the derivative to find $f'(x)$ for $f(x) = 5x - 4$.</p> <p>B4 Use the definition to find $f'(x)$ for $f(x) = x^2 + 3x$.</p> <p>B5 Find the equation of the tangent to $y = x^2$ at $x = 2$. Use $f'(x) = 2x$.</p> | <p>B6 For $f(x) = x^3$, find the secant slope between $x = 1$ and $x = 2$. Compare it with $f'(1) = 3$.</p> <p>B7 For $f(x) = x^2$ at $x = 2$, find the secant slopes for $h = 0.1, 0.01, 0.001$. What number are they approaching?</p> <p>B8 Use the definition to find $f'(x)$ for $f(x) = x^2 - 4x + 1$.</p> <p>B9 At which point on $y = x^2$ is the gradient equal to 10?</p> <p>B10 A car travels $s(t) = t^2$ metres in t seconds. Find its average speed from $t = 2$ to $t = 5$, and its speed at $t = 2$ given $s'(t) = 2t$.</p> |
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●●● Section C · Hard

Qiyin · Трудные

Algebra with letters, and questions that ask you to explain.

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| <p>C1 Use first principles to find $f'(x)$ for $f(x) = \frac{1}{x}$.</p> <p>C2 Use first principles to find $f'(x)$ for $f(x) = \sqrt{x}$.</p> <p>C3 Use first principles to find $f'(x)$ for $f(x) = x^3$.</p> <p>C4 Find the equation of the tangent to $y = x^2 - 2x$ at the point where the gradient is 4. Use $y' = 2x - 2$.</p> <p>C5 Show that the tangent to $y = x^2$ at $x = a$ has equation $y = 2ax - a^2$.</p> <p>C6 The tangent to $y = x^2$ at $x = a$ passes through (0, -4). Find a.</p> | <p>C7 Find the points on $y = x^3 - 3x$ where the tangent is horizontal. Use $y' = 3x^2 - 3$.</p> <p>C8 For $f(x) = x^2$, show that the secant slope between $x = a$ and $x = b$ is $a + b$. What happens as $b \rightarrow a$?</p> <p>C9 A ball has height $h(t) = 20t - 5t^2$ metres after t seconds. (a) Find its average velocity from $t = 1$ to $t = 3$. (b) Use first principles to find $h'(t)$. (c) When is the ball momentarily at rest?</p> <p>C10 For $f(x) = x$: the secant between $x = -1$ and $x = 1$ has slope 0. Explain why the graph still has no tangent at $x = 0$.</p> |
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●○○ Section A • Easy

A1 $\text{slope} = \frac{10-2}{5-1} = \frac{8}{4} = 2$ 2

A2 $y = 3x - 1 \rightarrow y = 3(4) - 1 = 11$ 11

A3 $\frac{5-5}{4-(-2)} = \frac{0}{6} = 0$ 0

A4 $\frac{16-1}{4-1} = \frac{15}{3} = 5$ 5

A5 $\frac{25-4}{5-2} = \frac{21}{3} = 7$ 7

A6 $\frac{19-4}{6-1} = \frac{15}{5} = 3$ $\rightarrow$ It equals the slope of the line itself. 3

Every secant of a straight line has the same slope — that is what "straight" means.

A7 $f(3) = 2(3) = 6$ 6

A8 $2x = 0$, so $x = 0 \rightarrow$ At the point $(0, 0)$ (0, 0)

A9 $\frac{(2+h)^2 - 4}{h} = \frac{4 + 4h + h^2 - 4}{h} = \frac{4h + h^2}{h} = 4 + h$ 4 + h

A10 The gradient of the tangent to the curve at $x = 4$ is 7. $\rightarrow$ Near that point the curve rises about 7 units of y for every 1 unit of x .

gradient 7 at $x = 4$

It is the steepness at one point, not an average over an interval.

●●○ Section B • Medium

B1 $\text{slope} = \frac{11-3}{6-2} = 2 \rightarrow y - 3 = 2(x - 2) \rightarrow y = 2x - 1$ $y = 2x - 1$

B2 $\frac{(3+h)^2 - 9}{h} = \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h$ 6 + h

B3 $\frac{5(x+h) - 4 - (5x - 4)}{h} = \frac{5h}{h} = 5 \rightarrow f'(x) = 5$ 5

B4 $\frac{(x+h)^2 + 3(x+h) - x^2 - 3x}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \rightarrow$
 $f'(x) = 2x + 3$ $2x + 3$

B5 gradient $= f'(2) = 4$, point $(2, 4) \rightarrow y - 4 = 4(x - 2) \rightarrow y = 4x - 4$
 $y = 4x - 4$

B6 $\frac{8-1}{2-1} = 7 \rightarrow$ The average gradient over the interval is 7, but the

gradient at the left end is only 3. 7 (against 3)

The curve gets steeper as x grows, so the average over $[1, 2]$ is much bigger than the value at $x = 1$.

B7 secant slope $= 4 + h \rightarrow 4.1, 4.01, 4.001 \rightarrow$ They approach 4, and $f'(2) = 4$. 4

B8 $\frac{(x+h)^2 - 4(x+h) + 1 - (x^2 - 4x + 1)}{h} = \frac{2xh + h^2 - 4h}{h} =$

$2x + h - 4 \rightarrow f'(x) = 2x - 4$ $2x - 4$

B9 $2x = 10$, so $x = 5 \rightarrow y = 5^2 = 25$, so the point is $(5, 25)$ (5, 25)

B10 average $= \frac{25-4}{5-2} = 7$ m/s $\rightarrow$ at $t = 2$: $s'(2) = 4$ m/s $\rightarrow$ The car is

speeding up, so the average over the interval is faster than the

speed at the start. 7 m/s; 4 m/s

●●● Section C • Hard

C1 $\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{\frac{x - (x+h)}{h \cdot x(x+h)}}{h} = \frac{-h}{h \cdot x(x+h)} = \frac{-1}{x(x+h)}$ $\rightarrow$ Let

$h \rightarrow 0$: $f'(x) = -\frac{1}{x^2}$ $-1/x^2$

C2 Multiply above and below by $\sqrt{x+h} + \sqrt{x} \rightarrow \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} =$

$\frac{1}{\sqrt{x+h} + \sqrt{x}} \rightarrow$ Let $h \rightarrow 0$: $f'(x) = \frac{1}{2\sqrt{x}}$ $1/(2\sqrt{x})$

C3 $\frac{(x+h)^3 - x^3}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \rightarrow$ Let

$h \rightarrow 0$: $f'(x) = 3x^2$ $3x^2$

C4 $2x - 2 = 4$, so $x = 3 \rightarrow y = 9 - 6 = 3$, so the point is $(3, 3) \rightarrow$
 $y - 3 = 4(x - 3) \rightarrow y = 4x - 9$ $y = 4x - 9$

C5 gradient $= 2a$, point $(a, a^2) \rightarrow y - a^2 = 2a(x - a) \rightarrow$
 $y = 2ax - 2a^2 + a^2 = 2ax - a^2$ shown

C6 Use question C5: $y = 2ax - a^2 \rightarrow$ Put $x = 0$: $-a^2 = -4$, so $a^2 = 4 \rightarrow$
 $a = 2$ or $a = -2$ $a = \pm 2$

C7 $3x^2 - 3 = 0 \rightarrow x^2 = 1 \rightarrow x = \pm 1 \rightarrow x = 1$: $y = 1 - 3 = -2 \rightarrow (1, -2)$
 $\rightarrow x = -1$: $y = -1 + 3 = 2 \rightarrow (-1, 2)$ (1, -2) and (-1, 2)

C8 $\frac{b^2 - a^2}{b - a} = \frac{(b - a)(b + a)}{b - a} = a + b \rightarrow$ As $b \rightarrow a$, the slope $\rightarrow 2a \rightarrow$
 So $f'(a) = 2a$ $a + b \rightarrow 2a$

The whole chain in one line: secant $\rightarrow$ tangent $\rightarrow$ derivative.

C9 (a) $h(1) = 15$, $h(3) = 60 - 45 = 15 \rightarrow$ average $= \frac{15 - 15}{3 - 1} = 0$ m/s

$\rightarrow$ (b) $\frac{20(t+k) - 5(t+k)^2 - (20t - 5t^2)}{k} = 20 - 10t - 5k \rightarrow$ Let

$k \rightarrow 0$: $h'(t) = 20 - 10t \rightarrow$ (c) $20 - 10t = 0 \rightarrow t = 2$ s

0 m/s; 20 - 10t; $t = 2$ s

At the top of the flight the velocity is 0.

C10 secant slope $= \frac{1-1}{1-(-1)} = 0 \rightarrow$ From the right ($h > 0$): $\frac{|h|}{h} = 1$.

From the left ($h < 0$): $\frac{|h|}{h} = -1 \rightarrow$ The two one-sided limits are

different, so the limit does not exist. no tangent — a corner

No single gradient at $x = 0$ — the graph has a sharp corner, and a corner has no tangent.